

Beginner Glossary – No Short Forms

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA

Rule for this course: We write the **full term** first. If a short form exists, we explain it in plain English. You do not need prior coding knowledge to read the PDFs.

General programming

Full term	Meaning in simple words
Programming language	A language humans use to give instructions to a computer (example: JavaScript).
Code / source code	The text you write that the computer runs.
Variable	A named box that stores a value (example: <code>let age = 25</code>).
Function	A reusable block of code that does one job (example: add two numbers).
Loop	Code that repeats (example: print numbers 1 to 100).
Conditional / if statement	Code that runs only when a condition is true.
Operator	A symbol that does something (<code>+</code> , <code>-</code> , <code>===</code> , <code>%</code>).
Modulo operator (<code>%</code>)	Gives the remainder after division. Example: <code>10 % 3</code> is <code>1</code> .
Syntax	The grammar rules of a language (where to put brackets, semicolons, etc.).
Runtime	When the program is actually running (not just being edited).
Compile / compilation	When the engine reads your code and prepares it before or during execution.
Error	The program stops or misbehaves because something is wrong.
Bug	A mistake in code that causes wrong behavior.
Debug	Find and fix bugs.

JavaScript-specific

Full term	Meaning in simple words
JavaScript	The programming language used in browsers and Node.js. We do not write "JS" alone in beginner sections.
Node.js	Lets you run JavaScript outside the browser (on your computer or server).
Browser	Chrome, Firefox, Safari – where websites run.
Console	A place to see printed output (<code>console.log</code>). In Node.js, your terminal is the console.
Read-Eval-Print Loop (REPL)	Interactive mode: you type one line, it runs immediately and prints the result. Node REPL = type <code>node</code> in terminal.
Hoisting	JavaScript moves declarations to the top of a scope during compilation. Behavior differs for <code>var</code> vs <code>let</code> / <code>const</code> .
Temporal Dead Zone	The time after a block starts but before a <code>let</code> or <code>const</code> line runs. Using the variable then causes ReferenceError . We do not write "TDZ" alone in beginner text.
ReferenceError	Error: you used a variable that does not exist yet or is not accessible.
TypeError	Error: you used a value in an invalid way (example: call something that is not a function).
Scope	Where a variable is visible (inside a function, inside <code>{ }</code> block, or globally).
Block	Code inside <code>{ }</code> and <code>}</code> curly braces.
Closure	A function that remembers variables from the place where it was created, even after the outer function finished.
<code>this</code> keyword	A special value that depends on how a function was called (not where it was written).
Type coercion	JavaScript automatically converts types (example: number <code>5</code> becomes string <code>"5"</code> when you use <code>+</code>

	with a string).
Strict equality (<code>===</code>)	Compares value and type (<code>"5" === 5</code> is false).
Loose equality (<code>==</code>)	Compares with coercion (<code>"5" == 5</code> is true). Prefer <code>===</code> .
Primitive type	Simple single values: string, number, boolean, undefined, null, symbol, bigint.
Reference type / object	Complex values stored by reference: plain objects, arrays, functions.
Array	Ordered list of values: <code>[1, 2, 3]</code> .
Object	Collection of key-value pairs: <code>{ name: "Amit", age: 25 }</code> .
Map	Built-in structure for key-value pairs with fast lookup (<code>new Map()</code>).
Set	Built-in structure that stores unique values only (<code>new Set()</code>).
Hash map	Another name for using an object or Map to store values by a key for fast lookup.
Callback function	A function passed as an argument to run later.
Promise	An object that represents work that will finish in the future (success or failure).
Async / await	Syntax to wait for slow work (network, file) without freezing the whole program.
Event loop	How JavaScript handles many tasks on one thread (timers, network, promises).
Document Object Model (DOM)	The browser's tree of HTML elements your JavaScript can change.
ECMAScript	The official standard/specification for JavaScript. "ECMAScript 2015" is the same as "ES6" (year 2015 update).
Mozilla Developer Network (MDN)	Free, trusted documentation website for web technologies.

Speed of algorithms (time complexity)

We explain these in **plain English** for beginners:

Symbol	Plain English	Example
Constant time	Same speed no matter how big the input is.	Looking up a value in a Map by key.
Linear time	Work grows in proportion to input size (100 items $\approx$ 100 steps).	One loop through an array.
Quadratic time	Nested loops – slow for large inputs.	Comparing every pair in an array.
Logarithmic time	Very efficient search in sorted data.	Binary search (later topic).

(In advanced sections you may see $O(n)$ notation — it always means "linear time" as above.)

Tools and websites

Full term	Meaning
LeetCode	Website with coding practice problems (we write "LeetCode problem 1", not "LC 1").
Git	Version control – saves history of your code changes.
GitHub	Website to host Git repositories online.
Visual Studio Code	Popular free code editor (we say full name or "code editor").
Terminal / command line	Text interface where you type commands like <code>node file.js</code> .
Application Programming Interface (API)	A way for programs to talk to each other (example: fetch weather data from a server).

Web stack (later months – preview only)

Full term	Meaning
HTML	HyperText Markup Language – structure of web pages.
CSS	Cascading Style Sheets – styling and layout.
React	JavaScript library for building user interfaces.
Express	Node.js framework for building web servers and APIs.
MongoDB	Document database (stores JSON-like documents).
PostgreSQL	Relational database (tables with rows and columns).
MySQL	Another relational database (very common in companies).
Redis	In-memory database used mainly for caching and speed.

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